

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

*I S. Likiteswar Reddy(192425120)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: S. Likiteswar Reddy

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	

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Course : MLA0404 - DEEP LEARNING

Tool : C01 – AT2 - Error Analysis

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Q1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Introduction

The Mean Squared Error (MSE) measures the average squared difference between the actual values and the predicted values.

$$\text{MSE} = (1 / n) \times \sum (\text{Actual Value} - \text{Predicted Value})^2$$

A lower MSE indicates better prediction accuracy.

In this problem:

- Training MSE = 0.85
- Testing MSE = 5.20

The testing error is much higher than the training error. This indicates that the neural network performs well on the training data but poorly on unseen data. This situation is commonly known as overfitting.

Error Identification

The large difference between the training and testing MSE suggests that the model has memorized the training data instead of learning general patterns.

$$\text{Difference in MSE} = 5.20 - 0.85 = 4.35$$

Since the testing error is significantly higher, the model is not generalizing well to new data.

Possible Reasons for the Difference

1. Overfitting

The model may have learned the training data too closely, including noise and unnecessary details. As a result, it performs well during training but poorly during testing.

2. Insufficient Training Data

If the training dataset is too small, the model cannot learn all possible patterns, leading to poor performance on unseen data.

3. High Model Complexity

A neural network with too many hidden layers or neurons may become overly complex and memorize the training samples instead of learning general features.

4. Poor Data Quality

The dataset may contain noisy data, missing values, incorrect labels, or outliers, which reduce the model's ability to generalize.

5. Improper Train-Test Split

If the training and testing datasets have different data distributions, the model may not perform well on the testing dataset.

6. Lack of Regularization

Without regularization techniques, the network can develop excessively large weights, increasing the risk of overfitting.

Error Analysis and Reasoning

The training MSE of 0.85 indicates that the model has learned the training samples effectively.

However, the testing MSE of 5.20 is much larger, indicating that the learned knowledge is not applicable to unseen data.

The large gap between the two errors confirms that the neural network has poor generalization ability.

Instead of learning meaningful relationships between input and output, the model has memorized the training examples.

Corrective Measures

To improve the model performance, the following corrective measures can be applied:

1. Use Regularization

Apply L1 or L2 Regularization to reduce overfitting by preventing excessively large weights.

2. Apply Dropout

Dropout randomly disables some neurons during training, forcing the network to learn more generalized features.

3. Increase Training Data

Using more training samples or applying data augmentation helps the model learn a wider variety of patterns.

4. Early Stopping

Stop the training process when the validation error begins to increase. This prevents the model from overfitting.

5. Simplify the Model

Reduce the number of hidden layers or neurons so that the model focuses on learning important patterns instead of memorizing the training data.

6. Improve Data Quality

Remove duplicate records, handle missing values, eliminate outliers, and correct incorrect labels before training the model.

7. Cross-Validation

Use K-Fold Cross-Validation to evaluate the model on multiple data splits, resulting in a more reliable estimate of performance.

Interpretation of Results

The training MSE is very low (0.85), showing that the model performs well on the training dataset.

However, the testing MSE (5.20) is much higher, indicating poor prediction accuracy on new data.

This large difference clearly shows that the model is overfitting.

After applying techniques such as regularization, dropout, early stopping, improving data quality, increasing training data, and simplifying the model, the testing MSE can be reduced, improving the model's ability to generalize to unseen data.

Conclusion

The neural network has a training MSE of 0.85 and a testing MSE of 5.20, indicating that it suffers from overfitting. The model has memorized the training data instead of learning generalized patterns. By applying corrective measures such as regularization, dropout, early stopping, better data preprocessing, cross-validation, and increasing the training data, the testing performance can be significantly improved while reducing the gap between the training and testing errors.